



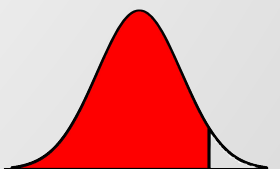
Decoding Market Movements: PCA-Enhanced Machine Learning for S&P 500 Log Return Prediction

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[Project Github](#)

Outline

1. Motivation & Introduction
2. Data
3. Benchmark model
4. Methodology
5. Result



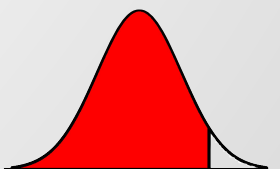
The challenge of S&P 500 forecasting

▣ Why S&P 500

- ▶ Market benchmark : proxies the U.S. economy
- ▶ Investment Impact: Core asset for portfolio management & hedging.

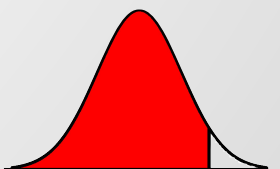
▣ Why is it hard

- ▶ Non-linearity: Complex interactions beyond linear models.
- ▶ High Noise (Chaos): Low signal-to-noise ratio.
- ▶ Dynamic Regimes: Structural breaks.



Research goal & contribution

- ▣ Previous work
 - ▶ Focus mostly on deep learning (high complexity, black box)
- ▣ Gap
 - ▶ Lack of systematic comparison between linear vs. non-linear feature engineering
- ▣ Objective
 - ▶ Target : predict daily log returns
 - ▶ Methodology : systematically evaluate 20 Model Configurations (4 Feature Sets * 5 Models)
 - ▶ Goal : compare models on both prediction accuracy and trading profitability



Data overview

- ▣ Source : Kaggle S&P 500 Volatility Prediction Time Series Data
- ▣ Time range : 2010/07/08 to 2019/10/04
- ▣ Frequency : Daily data
- ▣ Variable types
 - ▶ Index-related variables
 - ▶ Options market variables
 - ▶ Macroeconomic and financial indicators
 - ▶ Market sentiment variables
- ▣ Input (X_t) : 7-day lagged feature
- ▣ Target (Y_{t+1}) : Next-day log returns

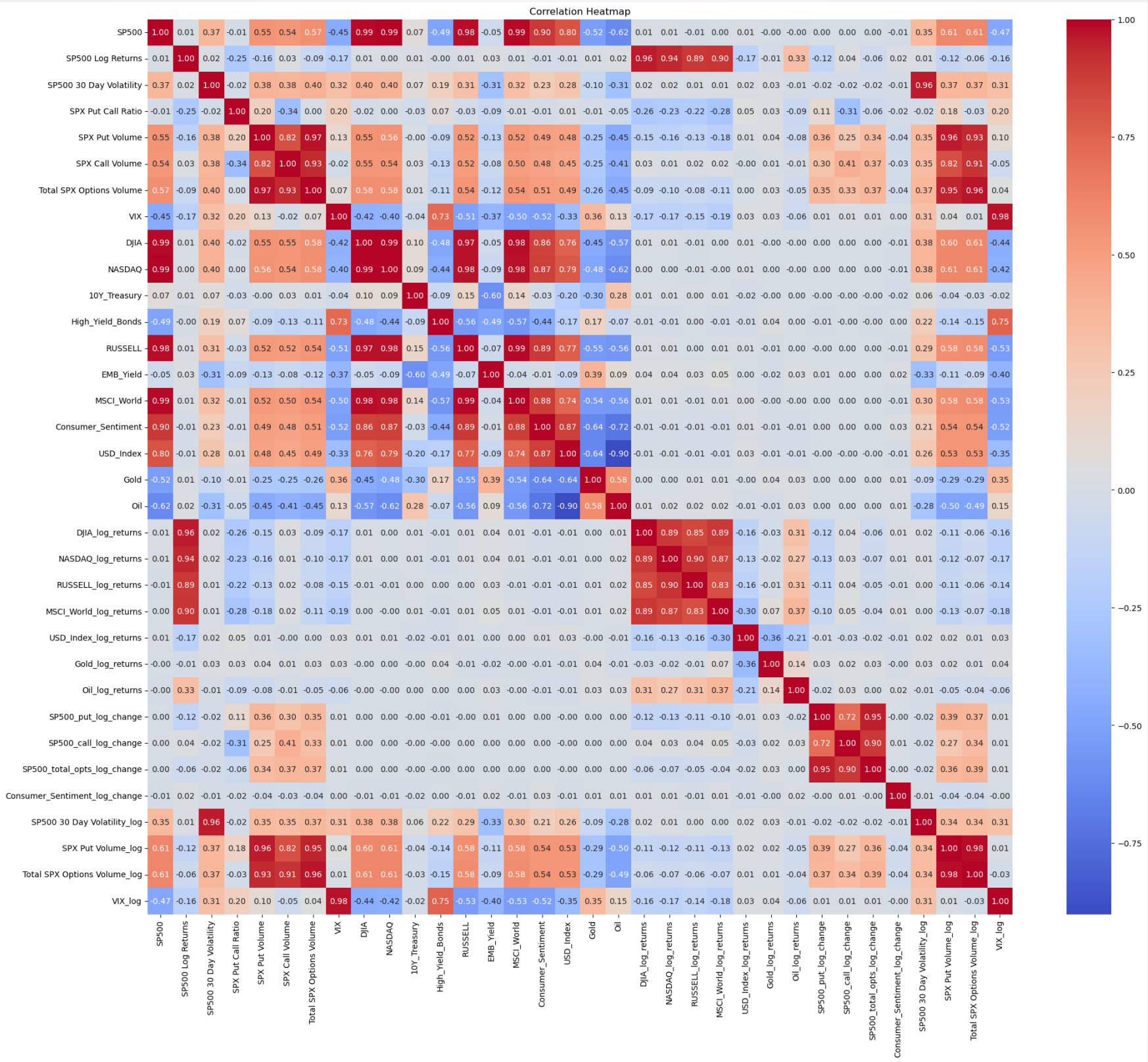


Data summary

	summary_statistics					
	mean	std	min	max	skewness	kurtosis
SP500	1978.67912	550.72266	1047.22000	3025.86000	0.14860	-1.10826
SP500 Log Returns	0.00046	0.00919	-0.06896	0.04840	-0.52531	4.82707
SP500 30 Day Volatility	31.39342	17.19281	8.03798	112.32049	1.70000	3.19323
SPX Put Call Ratio	1.74392	0.38484	0.81000	3.77000	0.58056	0.73029
SPX Put Volume	629725.68595	260946.40907	109844.00000	2312576.00000	1.38980	3.96059
SPX Call Volume	372806.37235	158016.18392	53326.00000	1036415.00000	0.99296	1.12755
Total SPX Options Volume	1002532.05830	400729.46757	163170.00000	3208442.00000	1.07055	1.91329
VIX	16.56075	5.36987	9.14000	48.00000	1.84294	4.68998
DJIA	17733.13310	4823.16764	9985.81000	27359.16000	0.42679	-0.94745
NASDAQ	4779.25528	1768.57388	2114.03000	8330.21000	0.39585	-1.02388
10Y_Treasury	2.34649	0.48964	1.36600	3.72500	0.28034	-0.57662
High_Yield_Bonds	6.67696	0.96561	5.16000	10.13000	0.93650	0.34330
RUSSELL	1148.55457	296.09852	595.59000	1740.75000	0.04797	-1.07539
EMB_Yield	111.66482	4.25524	102.36000	123.16000	0.50544	-0.14604
MSCI_World	1656.46321	316.64317	1068.50000	2248.89990	0.03494	-1.07359
Consumer_Sentiment	85.79271	11.21026	55.80000	101.40000	-0.56431	-0.70169
USD_Index	88.36747	8.41199	72.93000	103.25000	-0.07222	-1.57244
Gold	1355.67814	179.04747	1049.60000	1891.90000	0.92443	-0.09833
Oil	72.58707	22.58825	26.19000	113.39000	0.02465	-1.44833
DJIA_log_returns	0.00043	0.00876	-0.05706	0.04153	-0.56913	3.92336
NASDAQ_log_returns	0.00058	0.01070	-0.07149	0.05159	-0.51403	3.35888
RUSSELL_log_returns	0.00040	0.01223	-0.09332	0.06713	-0.36638	4.07421
MSCI_World_log_returns	0.00031	0.00826	-0.05261	0.04113	-0.57785	4.44935
USD_Index_log_returns	0.00007	0.00452	-0.02122	0.02469	0.09463	1.69704
Gold_log_returns	0.00010	0.01008	-0.09810	0.04611	-0.73115	7.32519
Oil_log_returns	-0.00015	0.02131	-0.09878	0.14176	0.19573	3.78394
SP500_put_log_change	0.00053	0.30787	-1.16060	1.03996	0.09621	0.24819
SP500_call_log_change	0.00012	0.35627	-1.22266	1.25620	0.05861	0.24228
SP500_total_opts_log_change	0.00037	0.30431	-1.14719	1.09183	0.11118	0.27022
Consumer_Sentiment_log_change	0.00015	0.00984	-0.13815	0.10077	-2.31619	80.71816



Heatmap



Linear regression on raw features

- ▣ OLS model
- ▣ Input (X_t) : 7-day lagged feature (210 features in total)
- ▣ Target (Y_{t+1}) : Next-day log returns

$$Y_{t+1} = \beta_0 + \sum_{i=1}^p \beta_i X_{t,i} + \epsilon_{t+1}$$

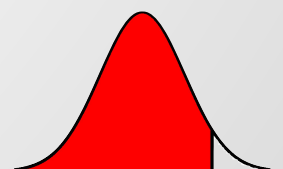


Benchmark results

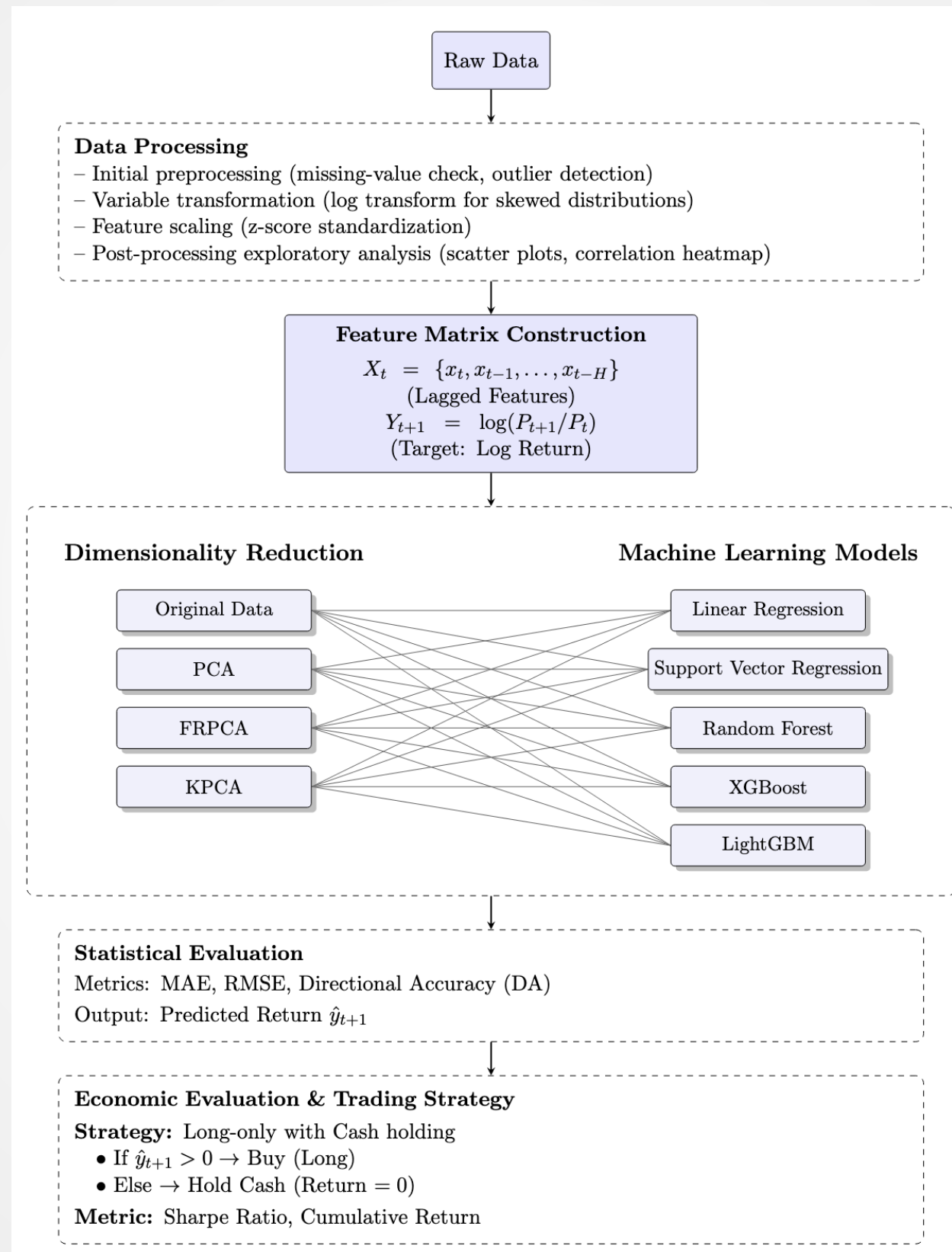
Omnibus:	202.800	Durbin-Watson:	1.980
Prob(Omnibus):	0.000	Jarque-Bera (JB):	492.126
Skew:	-0.527	Prob(JB):	1.37e-107
Kurtosis:	5.030	Cond. No.	1.41e+18

Evidence of Multicollinearity

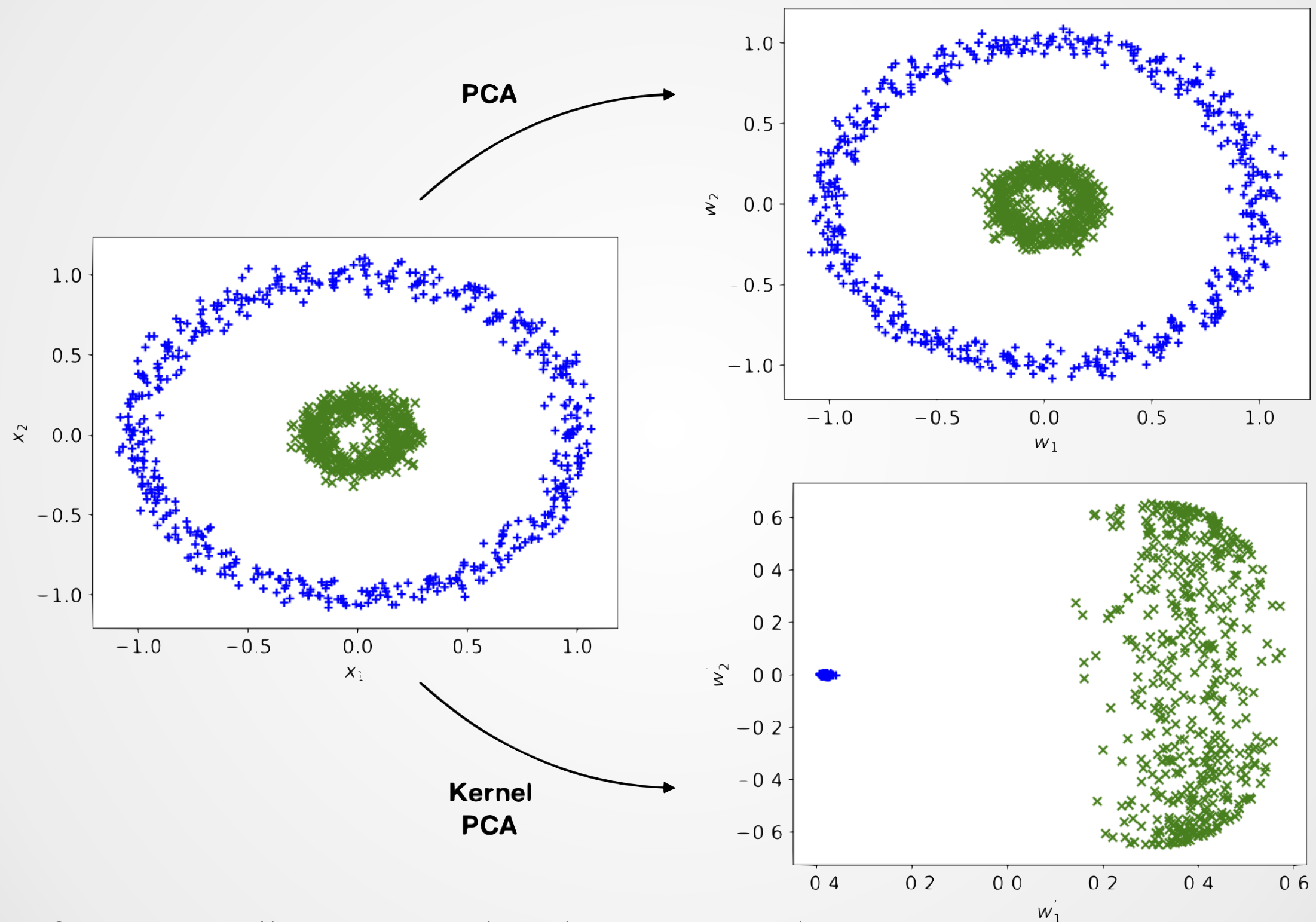
- Highly redundant information
- Metric
 - ▶ MAE : 0.01883
 - ▶ RMSE : 0.02262
 - ▶ DA : 0.4878



Study plan



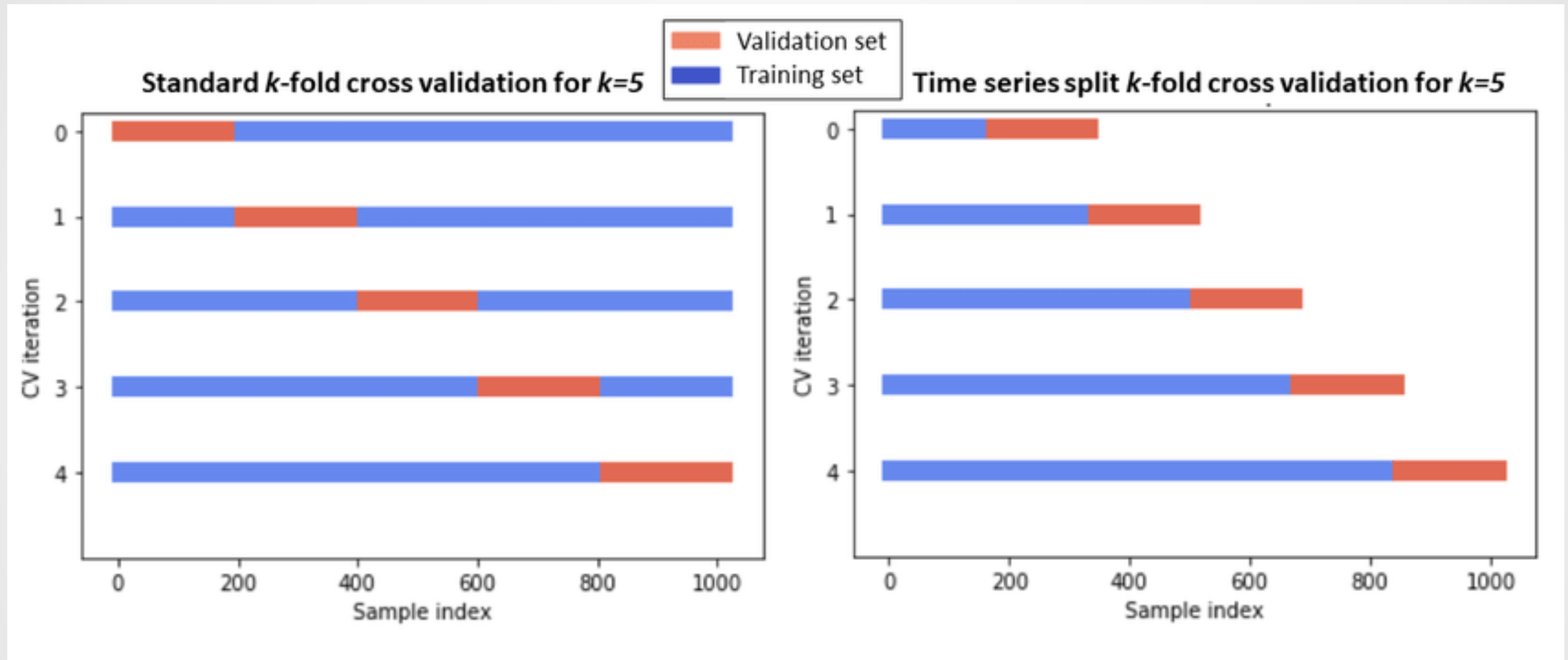
Methodology highlight : non-linear feature engineering



Source : https://ml-lectures.org/docs/structuring_data/ml_without_neural_network-2.html



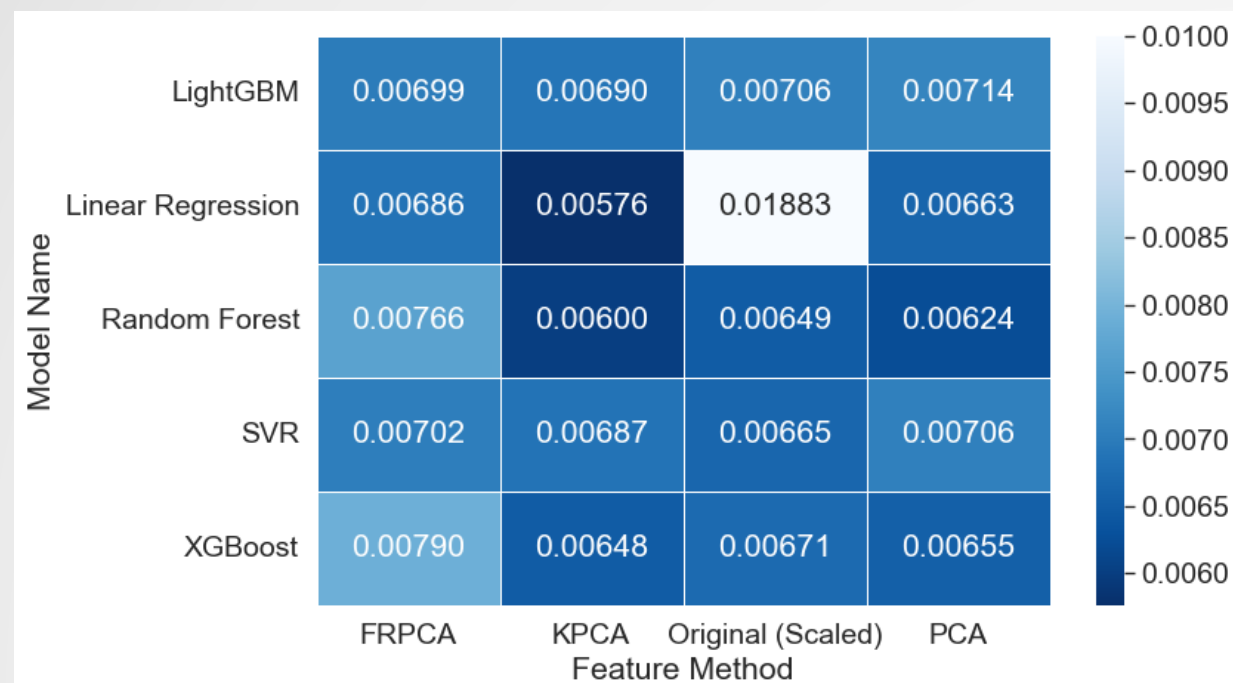
Methodology highlight : robust validation



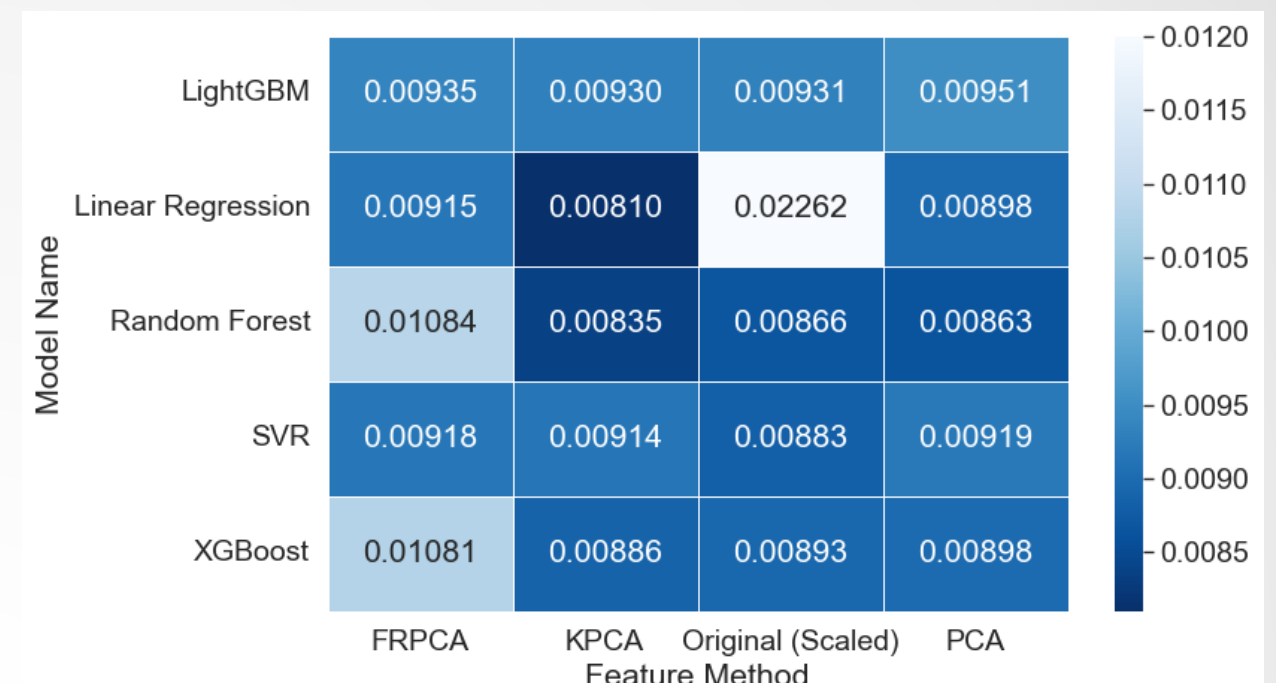
Source: https://www.researchgate.net/figure/Classical-k-fold-cross-validation-vs-time-series-split-cross-validation_fig1_355889701



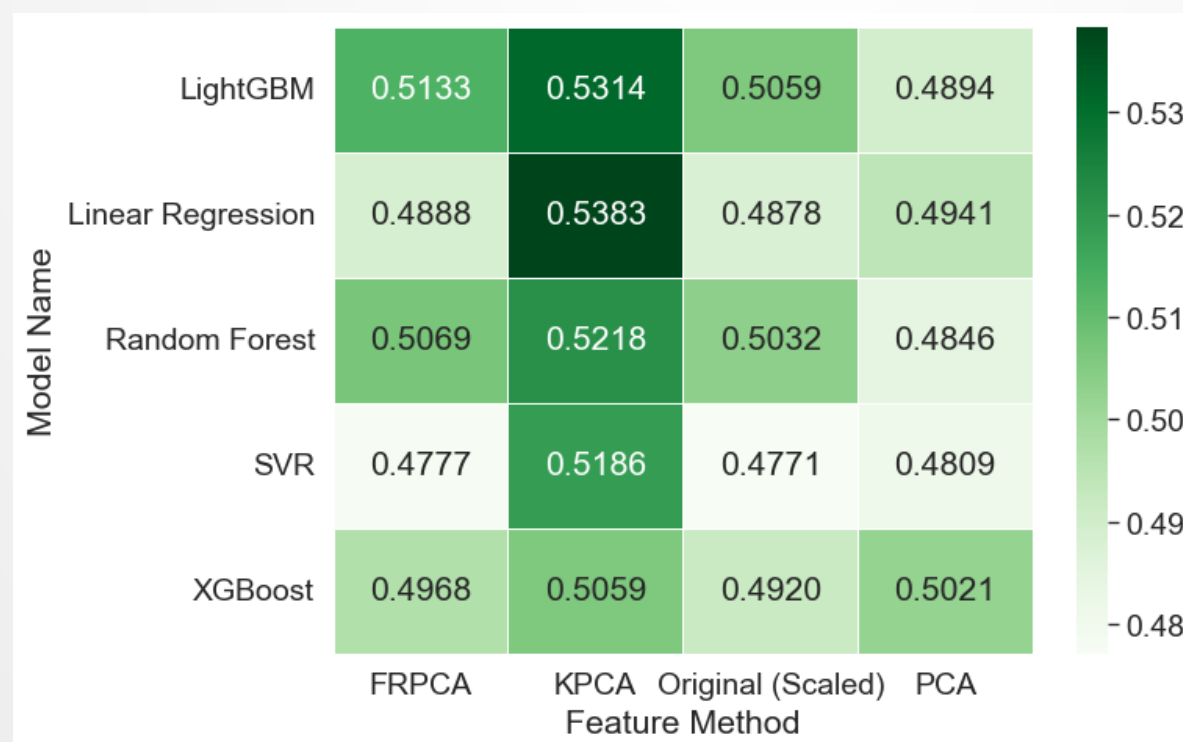
Statistical Results



(a) Test MAE (69% improvement)



(b) Test RMSE (64% improvement)



(c) Test DA



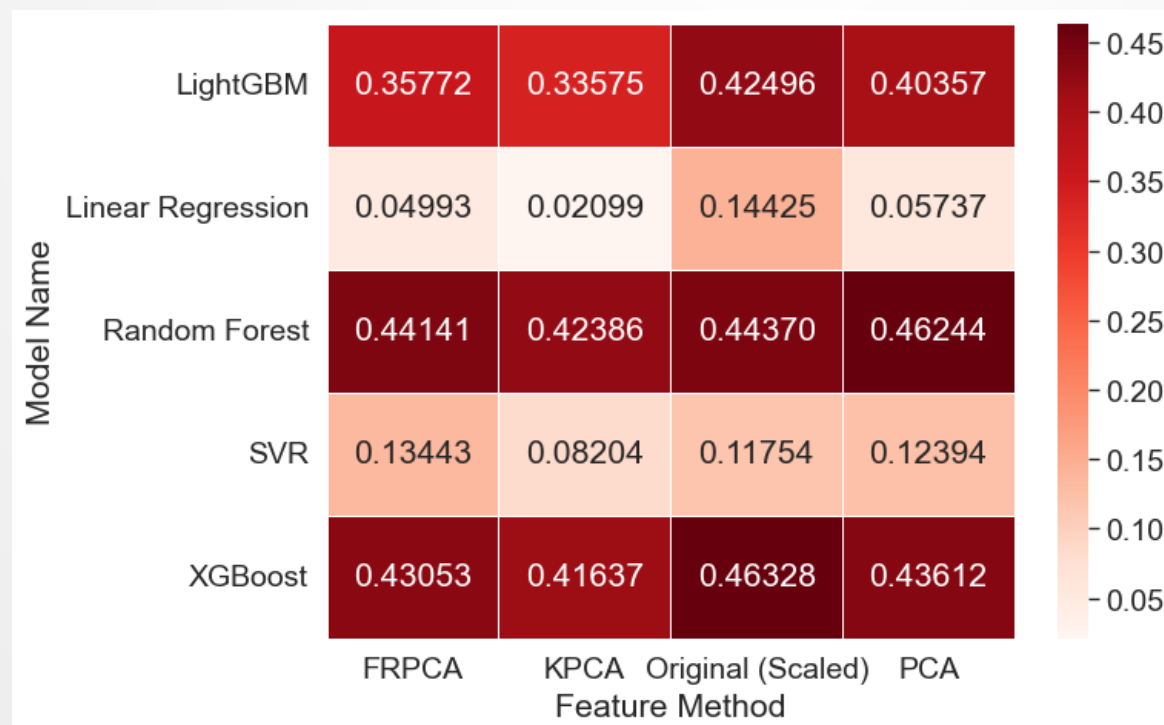
Robustness Analysis



(a) MAE gap



(b) RMSE gap



(c) DA gap



Economic Significance



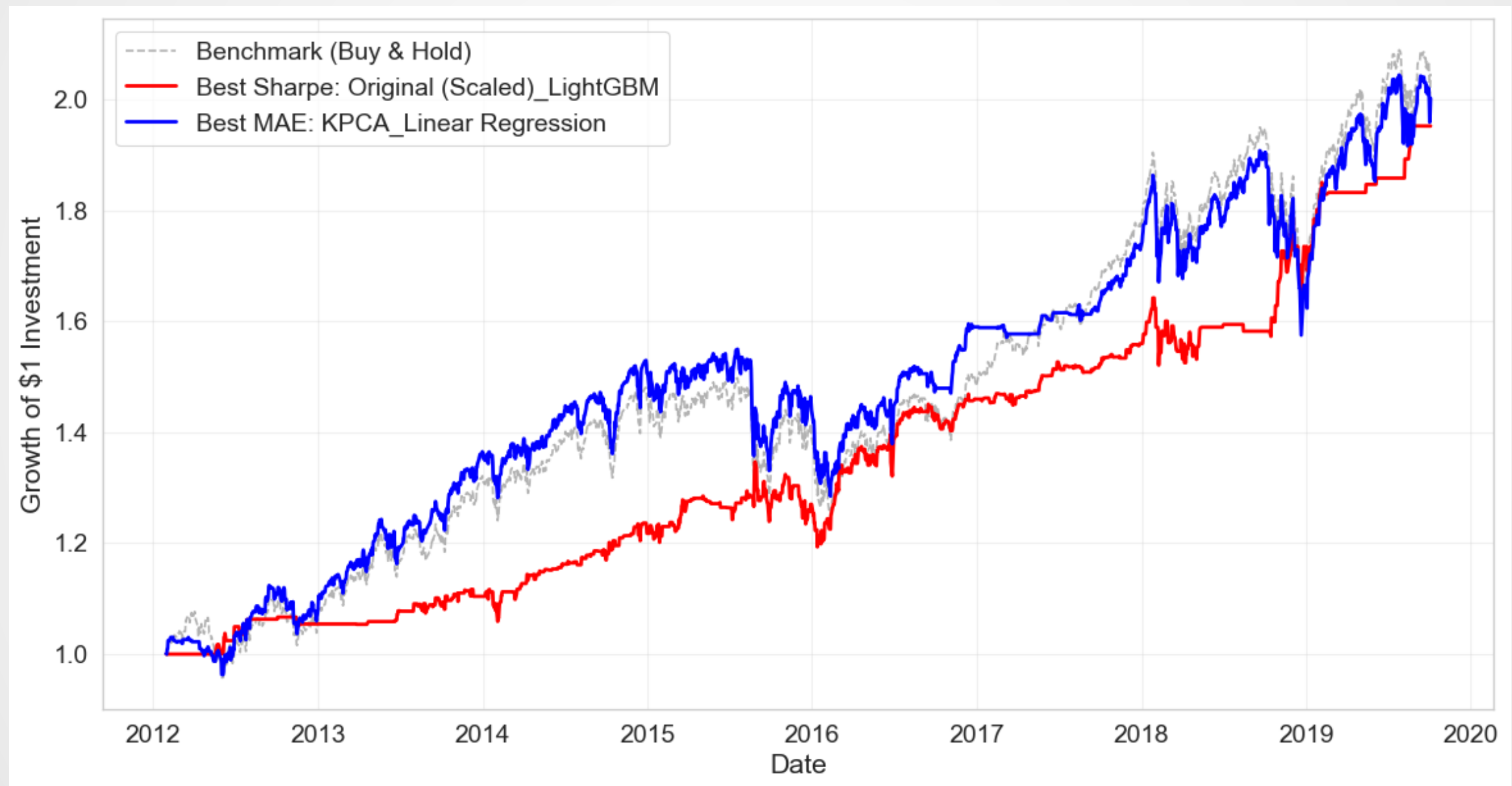
(a) Sharpe ratio

Rank	Feature Engineering	ML Model	Test MAE	Test RMSE	Test DA	Sharpe Ratio
1	Original (Scaled)	LightGBM	0.00706	0.00931	0.5059	1.116
2	KPCA	LightGBM	0.00690	0.00930	0.5314	0.881
3	KPCA	Random Forest	0.00600	0.00835	0.5218	0.873
4	FRPCA	LightGBM	0.00699	0.00935	0.5133	0.836
5	FRPCA	Random Forest	0.00766	0.01084	0.5069	0.830
-	<i>KPCA</i>	<i>Linear Regression*</i>	<i>0.00576</i>	<i>0.00810</i>	<i>0.5383</i>	<i>0.805</i>

(b) top 5 model ranked by Sharpe ratio



Economic Significance



Cumulative returns





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